

HIGH ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report was requested for High Acrylonitrile NBR, defined as nitrile butadiene rubber with elevated acrylonitrile (ACN) content, typically above standard NBR grades and often targeted for improved polarity and oil resistance. However, the provided patent manifest contains only one primary patent, US11813892B2, and its disclosure is not directed to NBR synthesis, acrylonitrile/butadiene copolymerization, or high-ACN composition control. Instead, the patent concerns acrylated/acylated or acetalized polyol monomers and their homopolymers, copolymers, block copolymers, and statistical copolymers prepared mainly by controlled radical polymerization, including RAFT, for applications such as asphalt modifiers, adhesives, tire compositions, and fracking fluids. The examples focus on glycerol-derived monomer preparation, esterification/acetalization chemistry, and free-radical polymerization of those monomers with initiators such as AIBN and chain-transfer agents such as CYCART in toluene at 80°C. No acrylonitrile, butadiene, NBR, emulsion polymerization, solution polymerization of nitrile rubber, or monomer-feed-ratio control relevant to high-ACN NBR is disclosed in the evidence. As a result, the patent is only indirectly relevant through its mention of tire compositions and polymerization process control concepts, but it does not provide actionable synthesis guidance for high acrylonitrile NBR. The evidence base therefore does not support a true high-ACN NBR patent comparison; the report captures the disclosed polymerization chemistry faithfully and flags the mismatch between the target compound and the available patent content.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11813892B2
- Patent Title: Acrylated and acylated or acetalized polyol as a biobased substitute for hard, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of acrylated and acylated or acetalized polyol monomers, followed by polymerization into homopolymers, copolymers, block copolymers, or statistical copolymers. The polymerizations are described as controlled radical processes, including RAFT, and some examples use free-radical polymerization with AIBN and a chain-transfer agent in solution. - 1-Glycerol methacrylate precursor: glycerol 1 mol; methacrylic acid 1.56 moles; Amberlyst 5 g per 100 mL of solution; 24 hours; 100° C — Example 2 - 1,3-Glycerol acetate precursor: glycerol 1 mol; acetic acid 10% mass by final solution mass; acetic anhydride 2 moles; Amberlyst 5 g per 100 mL of solution; approximately 16 hours — Example 3 - GRAMA v1: acetic anhydride 3 moles per mole of 1-glycerol methacrylate; room temperature; approximately 16 hours — Example 4 - GRAMA v2: methacrylic anhydride 2 moles per mole of 1,3-glycerol acetate; 80° C; 4 hours — Example 5 - Thermoset curing: varying weight percentages of AIBN; mold size 8.75 W x 11 L x 3.75 D; curing in a convection oven at varying temperatures; usually around 2-3 hours — Example 6 - Sub-functionalized precursor: glycerol 1 mol; methacrylic acid 0.91 moles; Amberlyst 5 g per 100 mL of solution; 24 hours; 100° C — Example 9 - Thermoplastics from GRAMA-v1: GRAMA 3 g; toluene 15 g; CYCART 0.006 g; AIBN 0.000493 g; argon 15 minutes; 80° C; 4 hours — Example 11 - SMA by Steglich esterification: methacrylic acid 10.33 g; dichloromethane 200 mL; DMAP 1.2 g; DCC 24.76 g; solketal 13.2 g; room temperature; 24 hours — Example 12 - SMA by transesterification: solketal 60 g; methyl methacrylate 135 g; potassium carbonate 3.1 g; phenothiazine 0.02 g; reflux over 5 Å molecular sieves — Example 13 - Thermoplastics of SMA: SMA 1 g; toluene 4 g; AIBN 0.00115 g; CYCART 0.00205 g; 80° C; 4 hours — Example 14 - Glycerol cyclopentanone ketal precursor: glycerol 119.6 g; cyclopentanone 84.12 g; p-toluene sulfonic acid 1.8 g; toluene as solvent; reflux — Example 15 - GCM precursor: glycerol-cyclopentanone ketal 154 g; excess MMA 300 g; 5 Å molecular sieves 230 g; Novozyme 435 6 g; 40° C; 24 hours — Example 16 - Thermoplastics of GCM: GCM 1 g; toluene 2 g; AIBN 0.00082 g; CYCART 0.00205 g; 80° C; 4 hours — Example 17 - Glycerol butanone ketal precursor: glycerol 157.5 g; butanone 370.43 g; p-toluene sulfonic acid 2.5 g; toluene as solvent; reflux — Example 18 - GMM precursor: glycerol-butanone ketal 100 g; excess MMA 228.3 g; 5 Å molecular sieves 151.5 g; Novozyme 435 4 g; 40° C; 24 hours — Example 19 - Thermoplastics of GMM: GCM 1 g; toluene 1 g; AIBN 0.00143 g; CYCART 0.00256 g; 80° C; 4 hours — Example 20

Relevant Experimental Evidence - Example 2: Demonstrated synthesis of 1-glycerol methacrylate from glycerol and methacrylic acid using Amberlyst at 100° C for 24 hours. - Example 4: Showed acetylation of 1-glycerol methacrylate to GRAMA-v1 with acetic anhydride at room temperature. - Example 5: Showed methacrylation of 1,3-glycerol acetate to GRAMA-v2 at 80° C for 4 hours. - Example 11: Demonstrated solution polymerization of GRAMA-v1 with AIBN and CYCART under argon at 80° C for 4 hours. - Example 14: Demonstrated polymerization of SMA with AIBN and CYCART at 80° C for 4 hours, yielding a glassy polymer.

Technical Relevance This patent is not a direct source for high acrylonitrile NBR synthesis because it contains no acrylonitrile/butadiene copolymerization disclosure. It is only tangentially relevant through its controlled radical polymerization methods and its mention of tire compositions, which may be of general interest for elastomer-adjacent materials design.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The provided evidence does not disclose a true high-acrylonitrile NBR patent set; the only patent in the manifest is unrelated polyol-based polymer chemistry. Therefore, no evidence-based synthesis parameters for high-ACN NBR can be recommended from this dataset. For actual high-acrylonitrile NBR development, the most relevant missing disclosures would be acrylonitrile/butadiene feed ratios, emulsion or solution polymerization conditions, initiator and chain-transfer systems, temperature control, and conversion/composition management.

5. REFERENCES

- US11813892B2 | Acrylated and acylated or acetalized polyol as a biobased substitute for hard, ... | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11813892B2/en>